

Dietary roles of non-starch polysaccharides in human nutrition: a review.



Nonstarch polysaccharides (NSPs) occur naturally in many foods. The physiochemical and biological properties of these compounds correspond to dietary fiber. Nonstarch polysaccharides show various physiological effects in the small and large intestine and therefore have important health implications for humans. The remarkable properties of dietary NSPs are water dispersibility, viscosity effect, bulk, and fermentability into short chain fatty acids (SCFAs). These features may lead to diminished risk of serious diet related diseases which are major problems in Western countries and are emerging in developing countries with greater affluence. These conditions include coronary heart disease, colo-rectal cancer, inflammatory bowel disease, breast cancer, tumor formation, mineral related abnormalities, and disordered laxation. Insoluble NSPs (cellulose and hemicellulose) are effective laxatives whereas soluble NSPs (especially mixed-link β -glucans) lower plasma cholesterol levels and help to normalize blood glucose and insulin levels, making these kinds of polysaccharides a part of dietary plans to treat cardiovascular diseases and Type 2 diabetes. Moreover, a major proportion of dietary NSPs escapes the small intestine nearly intact, and is fermented into SCFAs by commensal microflora present in the colon and cecum and promotes normal laxation. Short chain fatty acids have a number of health promoting effects and are particularly effective in promoting large bowel function. Certain NSPs through their fermented products may promote the growth of specific beneficial colonic bacteria which offer a prebiotic effect. Various modes of action of NSPs as therapeutic agent have been proposed in the present review. In addition, NSPs based films and coatings for packaging and wrapping are of commercial interest because they are compatible with several types of food products. However, much of the physiological and nutritional impact of NSPs and the mechanism involved is not fully understood and even the recommendation on the dose of different dietary NSPs intake among different age groups needs to be studied.